

INDIA.RUNS

Build what next India runs on

Track 01

AI Systems & Workflow Innovation Challenge

Team Name : S2

Problem Statement : **The AI Systems Architect: Reimagining Work**

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Problem Definition

Questions

- What problem are you solving?
 - Students and job seekers lack a structured, feedback-driven system to convert preparation into actual hiring outcomes. Preparation is fragmented (DSA, ML, projects), with no alignment to real job requirements.
- Who experiences this problem?
 - Students, fresh graduates, and early-career professionals using hiring platforms like Redrob.
- Why is the current approach insufficient?
 - Static courses and generic roadmaps
 - No real-time feedback or evaluation
 - No linkage between preparation and actual job requirements
 - No continuous improvement loop based on hiring outcomes

Opportunity & Vision

Questions

- Why is this an important opportunity?
 - India has millions of job seekers, but no system connects preparation directly with hiring success. Redrob already owns hiring workflows — adding intelligence here creates massive leverage.
- What future state are you enabling?
 - Enable a future where:
 - Every user has a personalized, adaptive preparation system
 - Preparation is dynamically aligned with real job requirements
 - Hiring success becomes predictable, measurable, and improvable

Solution Overview

Questions

- What is your proposed solution?
 - “Autonomous Career Copilot” — an AI-native preparation layer integrated within Redrob that continuously analyzes, plans, executes, and improves a user’s job readiness.
- What makes it AI-native rather than AI-assisted?
 - Not just recommendations → full workflow automation
 - Continuous decision-making loop (plan → act → evaluate → adapt)
 - Multi-agent system with autonomous reasoning
- Which existing Redrob capability, workflow, or user journey does your solution build upon?
 - Candidate profiles
 - Job listings and requirements
 - Recruiter interactions and feedback
 - Existing hiring workflows

User Journey / Workflow Diagram

Questions

- How does a user interact with the solution?
 - User accesses solution via Redrob platform (web/app)
 - Selects service or initiates a request
 - Provides inputs (data, preferences, task details)
 - Tracks progress and receives results
- How does information flow through the process?
 - Input captured → validated by system
 - Processed via AI / automation / rule engine
 - Routed to relevant modules/services
 - Output generated and delivered to user
 - Feedback collected for continuous improvement
- Where does the solution integrate with Redrob experiences?
 - User profile & authentication sync
 - Access to user history and prior workflows
 - Real-time status updates within Redrob dashboard
 - Output/results stored in Redrob ecosystem
 - Triggers for notifications, recommendations, and follow-ups

Mandatory Visual

- Workflow Diagram

User → Apply Job → AI Analysis → Prep Plan → Tasks → Evaluation → Feedback → Improved Applications

AI Logic & Decision Flow

Questions

- Where does AI intervene?
 - Resume and profile analysis
 - Skill gap detection
 - Task generation
 - Performance evaluation
 - Adaptive roadmap updates
- How are decisions made?
 - Input: Resume + Job Requirements
 - Model compares skill vectors
 - Generates gap score
 - Plans targeted learning path
 - Updates based on performance signals
- How do agents/models/reasoning systems interact?

Planner Agent → Task Agent → Evaluator Agent → Feedback Agent (closed loop)

Mandatory Visual

- AI Flow Diagram

Input → Analysis → Planning → Execution → Evaluation → Feedback Loop

System Architecture

Questions

- What components make up the system?
 - Frontend (User Dashboard)
 - Backend API Layer
 - Multi-Agent Orchestrator (LangGraph/CrewAI)
 - LLM Engine
 - Vector Database (FAISS/Chroma)
 - Evaluation Engine
- Which Redrob systems, workflows, or services are leveraged?
 - User profile APIs
 - Job database
 - Recruiter feedback signals
- How do services interact?

User → API → Agent System → LLM + DB → Output → Feedback loop

Mandatory Visual

- Architecture Diagram

User Layer → API → Agent Layer → AI Models + DB → Output

Data, Context & Intelligence Layer

Questions

- What data powers the solution?
 - User profiles
 - Job descriptions
 - Application history
 - Performance data
 - Recruiter feedbacks
- How is context retrieved, stored, or utilized?
 - Retrieved via embeddings (vector search)
 - Stored in vector DB
 - Continuously updated
- How does existing Redrob context, signals, or workflows improve the experience?
 - Real hiring data
 - Real recruiter expectations
 - Real outcome signals

Mandatory Visual

- Data Flow Diagram

Data Sources → Embedding → Vector DB → Retrieval → AI Decision

Scalability & Technical Feasibility

Questions

- How would this be implemented?
 - MVP: Resume parser + LLM planner + basic evaluator
 - Scaled: Multi-agent orchestration + real-time feedback
- How does the system scale?
 - Stateless APIs
 - Distributed agent execution
 - Vector DB scaling
 - Cloud-native deployment
- What technical challenges exist?
 - Accurate evaluation of user performance
 - Maintaining personalization at scale
 - Data privacy and security

Redrob Ecosystem Integration

Questions

- Which existing Redrob capabilities are being leveraged?
 - Hiring workflows
 - Candidate profiles
 - Job matching system
 - Recruiter interactions
- What new capability does your solution introduce?
 - AI-driven preparation intelligence layer
- How does it strengthen the overall Redrob ecosystem?
 - Higher user engagement
 - Better candidate quality
 - Improved hiring success rate
- What additional opportunities become possible after implementation?
 - AI recruiter assistant
 - Skill-based hiring engine
 - Predictive hiring analytics

Mandatory Visual

- Ecosystem Integration Diagram

Users ↔ AI Prep Layer ↔ Jobs ↔ Recruiters ↔ Feedback Loop

Impact & Success Metrics

Questions

- What measurable outcomes are expected?
 - Increased interview success rate
 - Reduced preparation time
 - Higher job match accuracy
- How will success be tracked?
 - Conversion rate (apply → interview → offer)
 - User engagement (daily active usage)
 - Skill improvement scores
 - Recruiter satisfaction
- What value is created for users and Redrob?
 - Users: structured, outcome-driven preparation
 - Redrob: stronger ecosystem and retention

Future Roadmap

Questions

- How could this evolve over 2–3 years?
 - Advanced AI mock interviews (voice + behavioral)
 - Real-time coding evaluation
 - Personalized career paths
- What future capabilities can be unlocked?
 - Fully autonomous career agent
 - Predictive hiring recommendations
 - Cross-platform AI career ecosystem
- What broader vision does this support?
 - A unified AI system where preparation, evaluation, and hiring are seamlessly connected — making career success data-driven and predictable.



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THANK YOU